



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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September 8, 2026



Outline

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2. Introduction
3. Methodology
4. Results
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Executive Summary

Business Understanding

Determine **first stage booster** recovery success using SpaceX Falcon 9 launches public data to **assess reusability** and estimate launch **cost advantages** based on first stage landing status as the **major cost driver**.

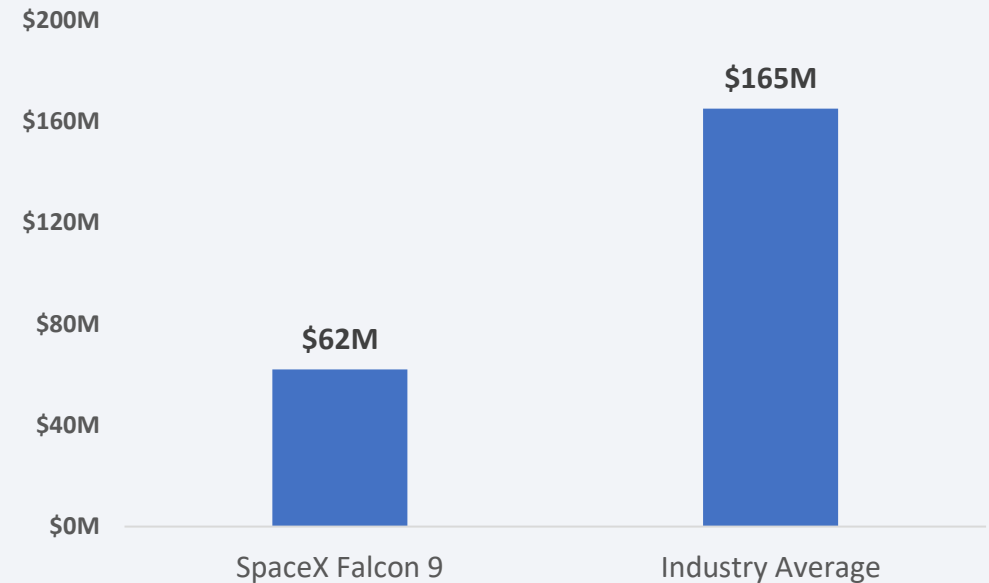
Core Findings

Our model achieved **83.33% test accuracy** within 18 samples, with predictive performance primarily explained by key operational hardware readiness and booster maturity i.e. **Legs**, **ReusedCount**, **GridFins** accounting for roughly **66% of feature importance**.

Recommendation:

Deploy **Logistic Regression** as the predictive model, it matches the performance ceiling of complex alternatives while delivering interpretable coefficients and calibrated recovery probabilities essential for commercial risk and pricing decisions.

SpaceX Launch Cost Advantages



Introduction

SpaceX advertises Falcon 9 rocket launches on its website with a cost of \$62 million; other providers cost upwards of \$165 million each, much of the savings is because **SpaceX can reuse the first stage**. Therefore, if we can **determine if the first stage will land**, we can determine the cost of a launch.

Project Background

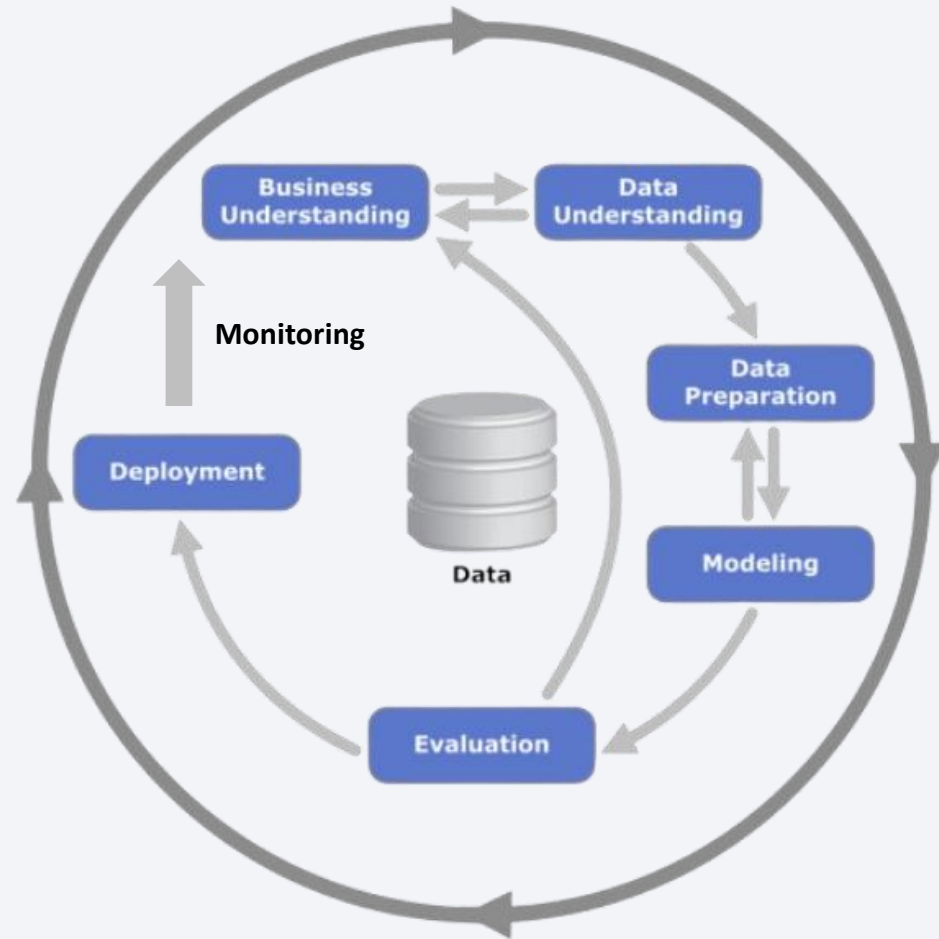
Assuming we would like learn about SpaceX Falcon 9 and determine the price of each launch, we will start by gathering information about SpaceX Falcon 9 Launch and then **determine** SpaceX **first stage** booster **reusability** after launch.

In this project, instead of using rocket science to predict whether SpaceX first stage will land successfully, we will try simple **machine learning approach** and make **use** of **public information**.

Problem Statement

1. What features or variables have relationship and able to determine whether first stage will land successfully?
2. Which machine learning model performs the best to predict whether first stage will land successfully?
3. How are the model performance evaluation on unseen blind test data?

Methodology



As standard guide through data analytics project, we adopt a framework known as **CRISP-DM** to structure each project phase.

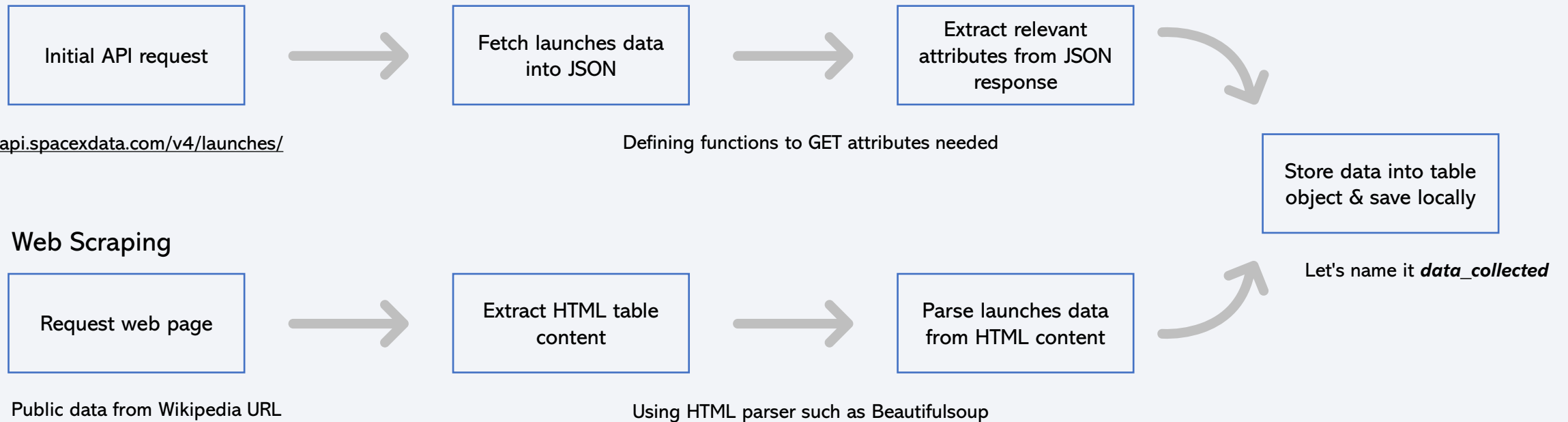
In this project, we will mainly cover:

1. Data Understanding
2. Data Preparation
3. Simple Model Training and Evaluation
4. Fraction of Model Deployment (demo purposes)

Data Understanding | Data Collection

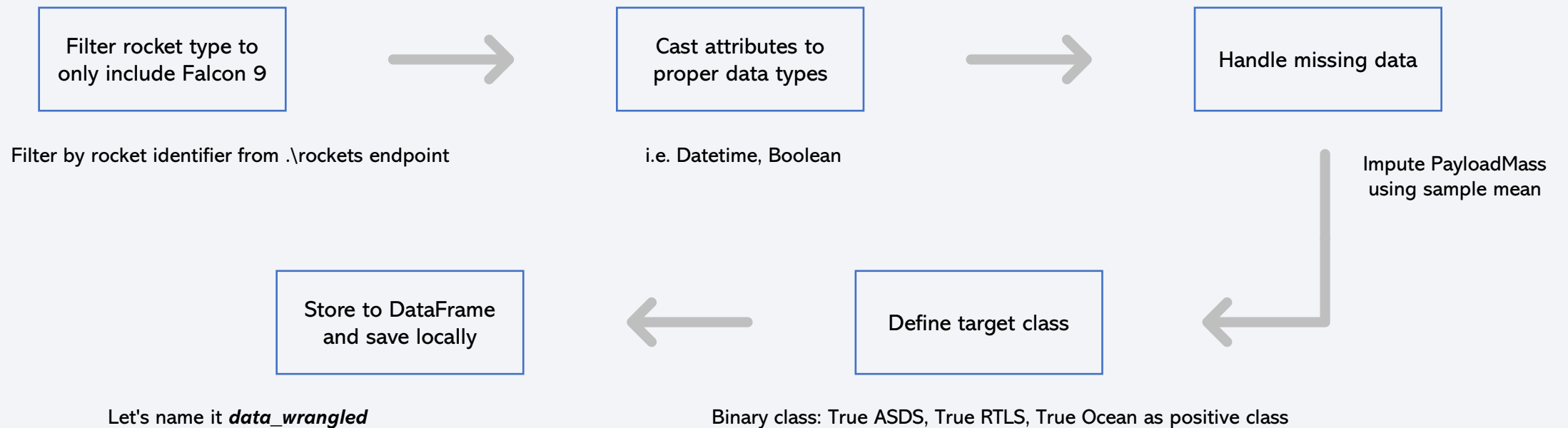
We can gather historical SpaceX Falcon 9 launches data using two different methods, using provided SpaceX **REST API** endpoint | **web scraping** and parsing from Wikipedia.

SpaceX API



Data Understanding | Data Wrangling

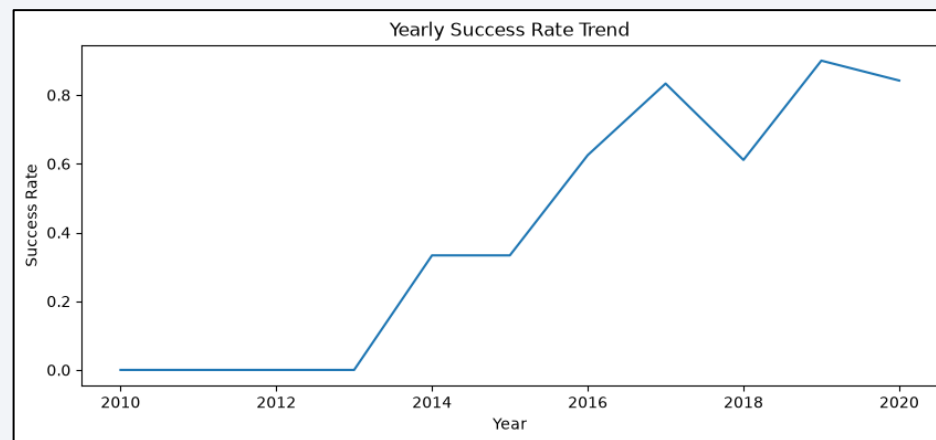
Raw launches records from *data_collected* need to be cleaned, validated, transformed into a **structured data format**.
Data wrangling process consists of resolving missing value or inconsistent format, filtering valid rocket type, and transforming launch outcomes as our target class.



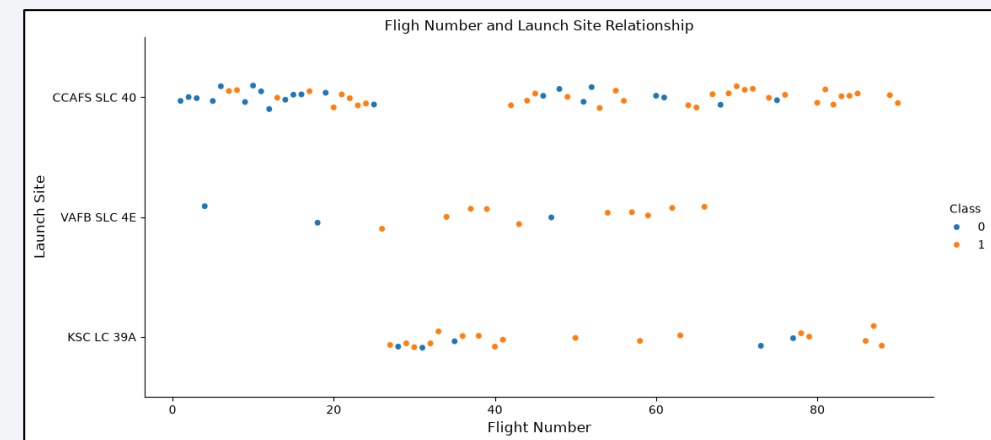
Data Understanding | EDA

Exploratory Data Analysis was conducted using data visualization to uncover early **insights and pattern** in flight number, launch site, payload mass, orbital and yearly success rates to first stage landing class. Recall that our target are first stage **success landing (Class = 1)**, first stage **fail landing (Class = 0)**.

Sample of Data Visualization



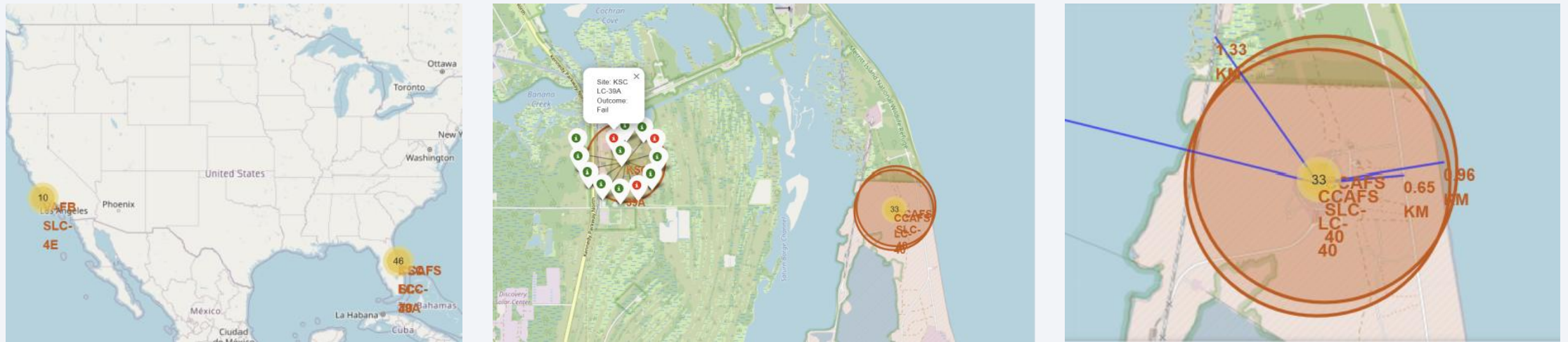
We can observe that the success rate since 2013 kept increasing till 2017, whereas it was stable in 2014 and then after 2015 it started increasing.



As flight number increases, the first stage is more likely to land successfully, and it seems the majority of first stage success landing are on VAFB and KSC, that might indicate more mature project launch.

Data Understanding | Proximities Analysis (Folium)

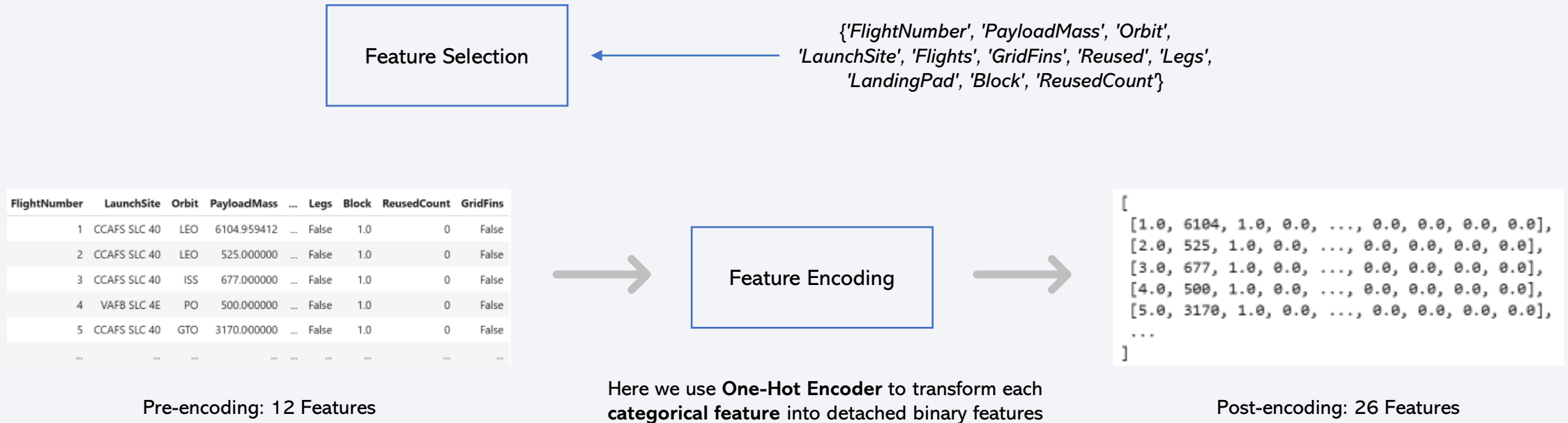
This section serves as an **additional** lookup and domain context for this project, working with **geospatial analysis**. Our interest is to **learn the pattern** behind launch sites, any spatial constraints, or landing location profile.



All launch sites share a strict operational footprint: positioned along coastlines (< 2 km) to launch safely over open ocean, buffered from cities (> 20 km) for public safety, and connected to transit corridors (1 – 3 km). This tight link between site location and rocket launch supports Launch Site is not a random predictor variable and should be included as feature on predictive modeling moving forward.

Data Preparation | Feature Engineering

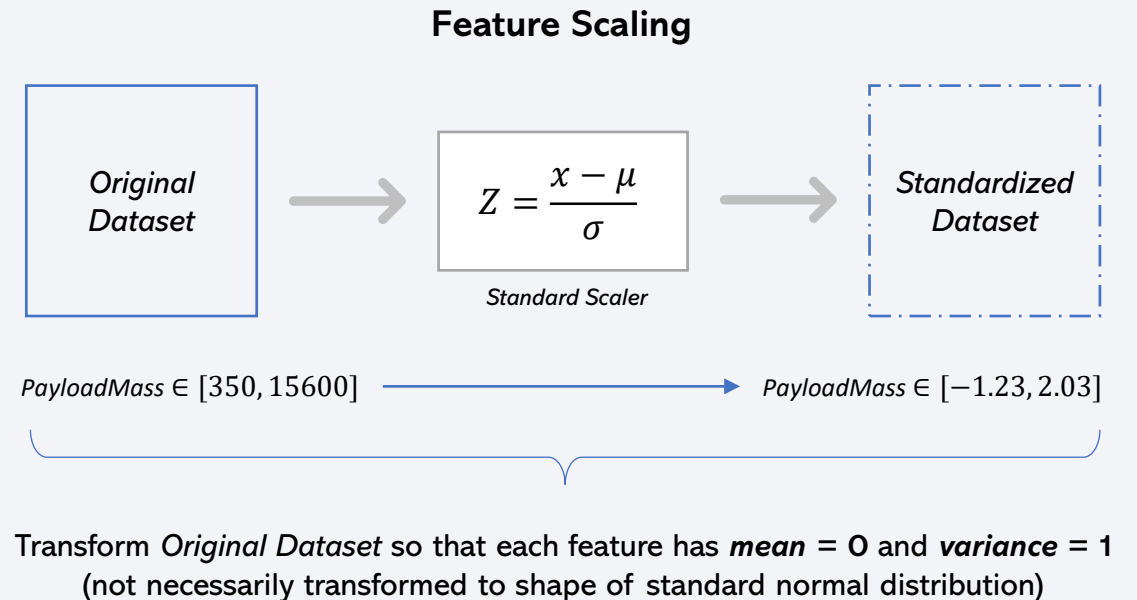
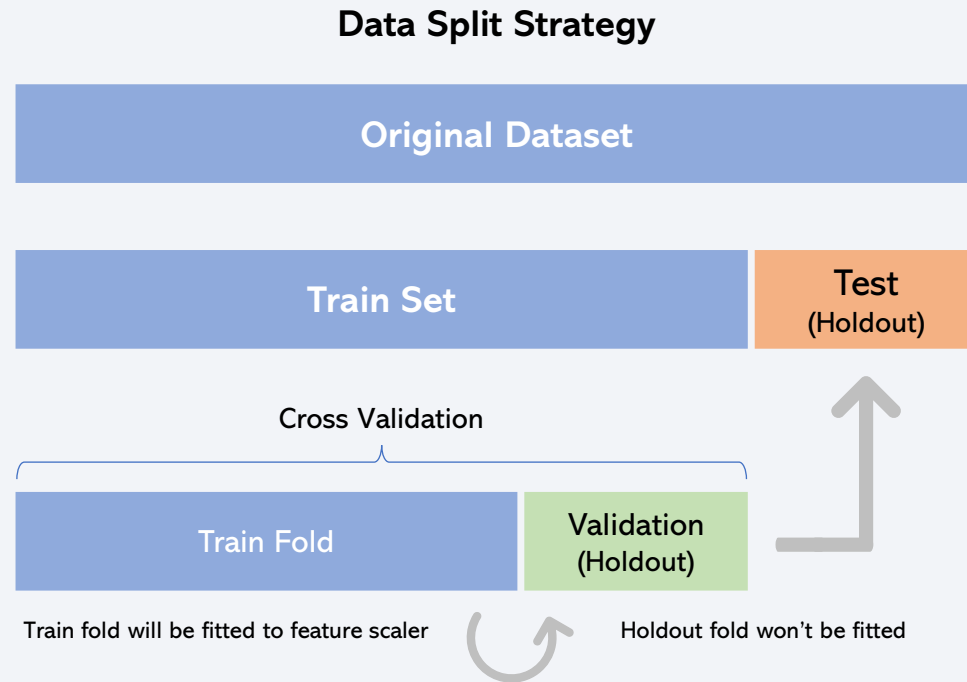
Prior to modeling process, we performed feature engineering on *data_wrangled* to craft tailored inputs (features) as predictor or independent variable for model training. Key steps made here are feature selection and feature encoding.



Let's make alias for current data as *data_prepared*

Data Preparation | Preprocessing

The 80-feature matrix from *data_prepared* was preprocessed through train/test splitting and feature scaling. Feature scaling is important to avoid unfair feature dominance and convergence, especially on linear (e.g. Logistic Regression) and distance-based (e.g. KNN, SVM) algorithms

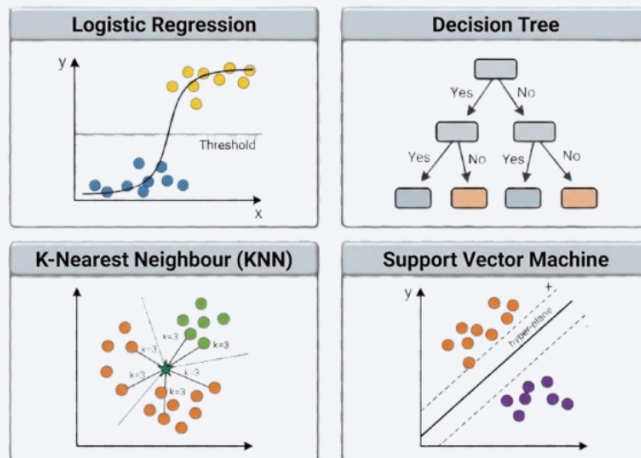


Modeling | Training & Evaluation

Since we have our target label, we then trained supervised learning model using **classification algorithms**. Model **hyperparameters** were tuned on 10-fold **Cross Validation** to evaluate generalization performance.

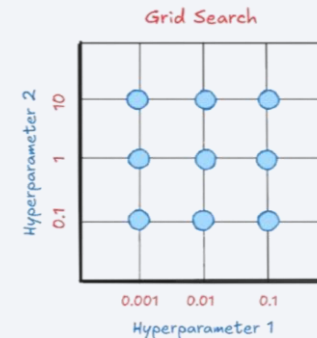
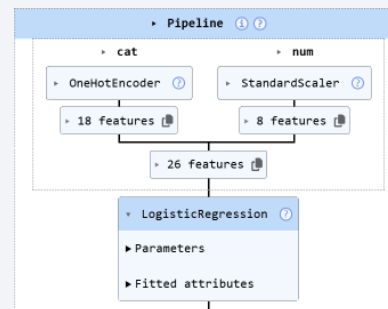
Training Algorithms

Model trained on these algorithms



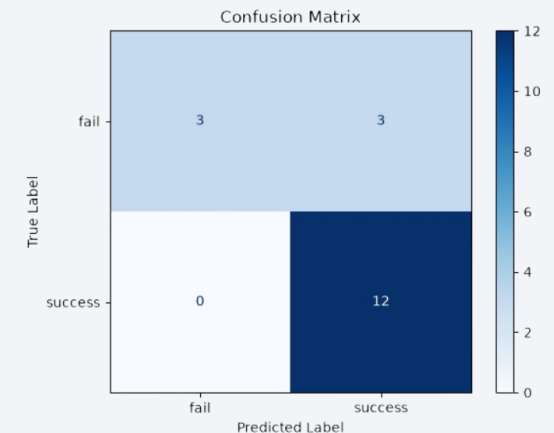
Model Tuning

Pipeline was used to streamline GridSearchCV tuning and prevent feature scaling leakage during CV



Model Evaluation

All model performed similarly with **weighted-recall score = 0.83**

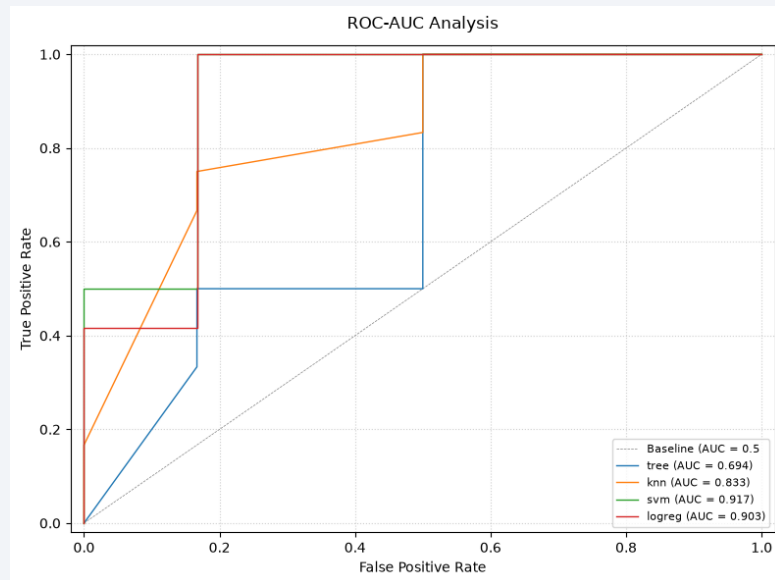


*We prioritize **recall** as main metrics to **minimize False Negative** case*

Results

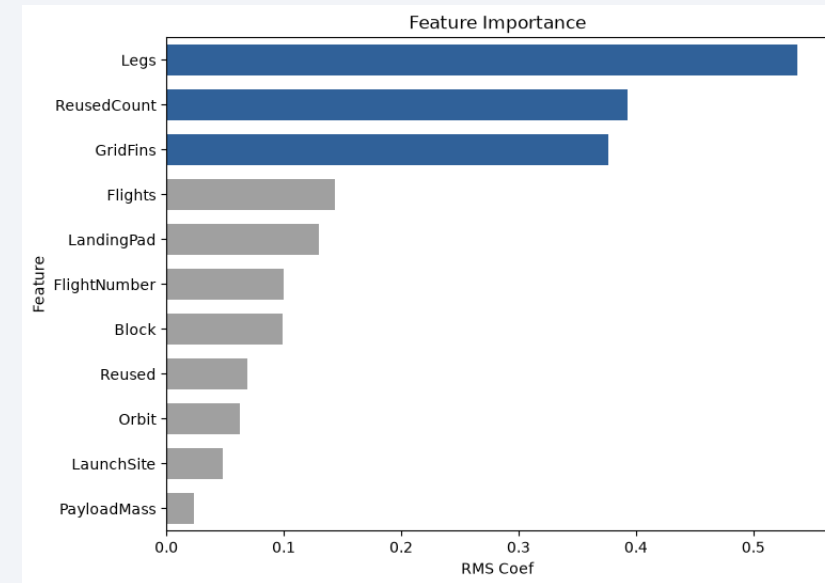
Let us assume linear decision boundary is acceptable to capture early phase of launch pattern.
In this case, we would suggest to select **LogisticRegression** as predictive model.

Model Selection



LogReg was selected as final model, supported by ROC-AUC and its class probabilities inference makes it easier to interpret.

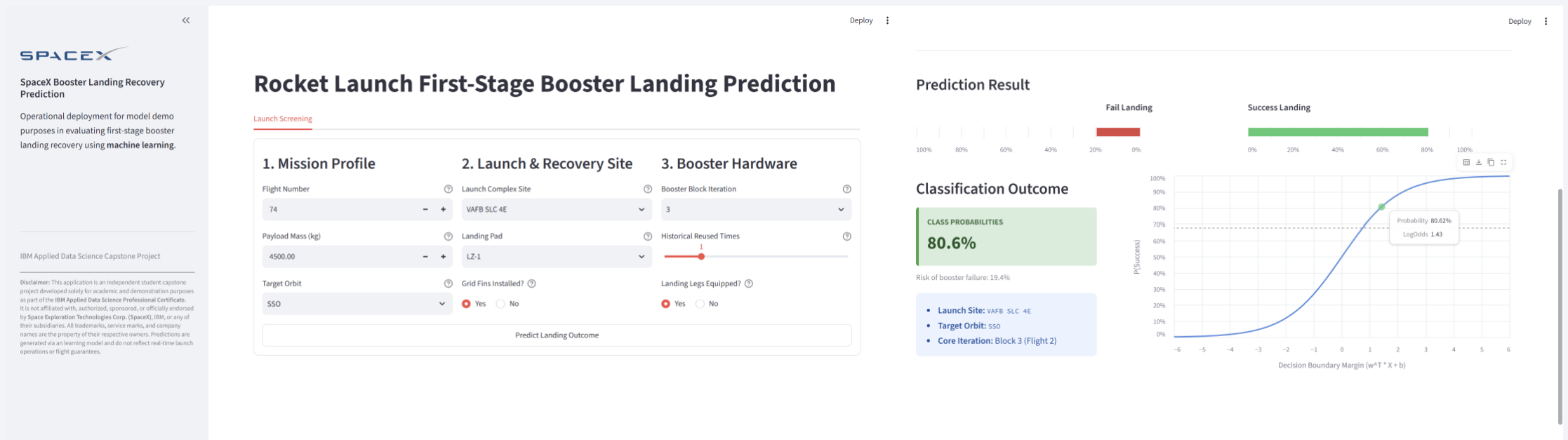
LogReg Feature Importance



We also analyzed the model feature importance using RMS coefficient, with Legs, ReusedCount, GridFins as the top-3 features.

Results

As for model demo purposes, we deployed our model into shareable Streamlit web app at:
<https://ibm-capstone-spacex-landing-prediction.streamlit.app/>



Conclusions

The simple machine learning models demonstrate solid predictive generalization using public launch data, reaching the practical limit of accuracy attainable without low-level descent telemetry and real-time metrics.

1

Based on our analysis using LogReg root mean squared coefficients, we can conclude that Legs, ReusedCount, and GridFins are top-3 features for LogReg model to determine first landing outcome. (not necessarily implying causality)

2

All models trained achieved identical test accuracy of 83.33%, that is 15 out 18 test case are correctly predicted. Moreover, the classification report are exactly identical (might indicate small sample for training).

3

Despite the similar performance of each models trained, we decided that Logistic Regression model is the best model to our case, having simple inference using probabilities and highly interpretable.

Appendix

1. Repository and source code: <https://github.com/Fery-K/spacex-falcon9-landing.git>
2. Deployed model demo: <https://ibm-capstone-spacex-landing-prediction.streamlit.app>
3. SpaceX REST API v4: <https://api.spacexdata.com/v4/launches>
4. Scikit-learn documentation: https://scikit-learn.org/stable/getting_started.html
5. Streamlit API references: <https://docs.streamlit.io/develop/api-reference>
6. Vega-Altair visualization library: <https://altair-viz.github.io/>(<https://altair-viz.github.io>

Thank you!

